# 30-Minute Student Task

Build Your Own Rule-Based Python Chatbot — Solved

Based on the requirements and target flow in the provided task sheet.

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## 1. Task Summary

The task is to build a simple rule-based conversational chatbot that continuously interacts with the user, recognizes different messages, gives appropriate responses, and exits when the user types “bye” or “exit.” The required flow is: INPUT → PROCESS → RESPOND → REPEAT → EXIT.

## 2. Requirement Checklist

| Requirement | How it is satisfied |
|---|---|
| Ask the user's name before the main conversation. | Implemented with input() before the loop. |
| Use while True. | Used to keep the conversation running. |
| Use input() to receive messages. | Used for the user's name and every message. |
| Clean input using .lower().strip(). | Applied to every conversation message. |
| Recognize at least two greetings. | Recognizes hello and hi, including variations using in. |
| Answer at least three questions/topics. | Covers Python, weather, help, and time-related topics. |
| Use if / elif / else. | The chatbot uses an if/elif/else decision structure. |
| Use break for bye or exit. | The loop breaks when the cleaned input is bye or exit. |
| Use the user's name in at least one response. | The greeting and goodbye use the user's name. |
| Useful response for unknown messages using else. | Unknown input receives a helpful fallback response. |
| Challenge: use in operator for flexible greetings. | Recognizes phrases such as 'hello bot' and 'hey hello'. |

## 3. Complete Python Solution

```
print("Welcome to the Rule-Based Python Chatbot!")
name = input("What is your name? ").strip()
print(f"Nice to meet you, {name}!")
print("You can say hello, ask about Python, ask about the weather,")
print("ask for help, or type 'bye' or 'exit' to leave.")
while True:
message = input("You: ").lower().strip()
if message == "bye" or message == "exit":
print(f"Goodbye, {name}! Have a great day.")
break
elif "hello" in message or "hi" in message:
print(f"Hello, {name}! How can I help you?")
elif "python" in message:
print("Python is a beginner-friendly programming language used for web development,
automation, data analysis, and AI.")
elif "weather" in message:
print("I cannot check live weather, but you can use a weather app or website for the current
forecast.")
elif "help" in message:
print("You can greet me, ask about Python, ask about the weather, or type 'bye' or 'exit' to end
the chat.")
elif "time" in message:
print("I cannot access your local clock in this simple rule-based chatbot.")
else:
print("Sorry, I do not understand that yet. Try saying hello, asking about Python or weather,
asking for help, or typing 'bye' to exit.")
```

## 4. How the Chatbot Works

**INPUT:** The program first asks the user for a name, then repeatedly uses input() to receive messages.

**PROCESS:** Each message is cleaned with .lower().strip(), so inputs such as HELLO and hello are treated consistently.

**RESPOND:** The if/elif/else statements compare the cleaned message with known rules and print the appropriate response.

**REPEAT:** while True keeps the conversation running after each response.

**EXIT:** If the user enters bye or exit, the chatbot prints a goodbye message and uses break to leave the loop.

## 5. Test Cases and Expected Results

| Test Input | Expected Result |
|---|---|
| Name Geethika | Nice to meet you, K. Bhargava Ram! |
| Greeting hello | Hello, K. Bhargava Ram! How can I help you? |
| Uppercase greeting HELLO | Hello, K. Bhargava Ram! How can I help you? |
| Second greeting hi | Hello, K. Bhargava Ram! How can I help you? |
| Flexible greeting hello bot | Hello, K. Bhargava Ram! How can I help you? |
| Flexible greeting hey hello | Hello, K. Bhargava Ram! How can I help you? |
| Question/topic What is Python? | A short explanation of Python is displayed. |

| Test Input | Expected Result |
| --- | --- |
| Question/topic Tell me the weather | The chatbot explains that it cannot check live weather. |
| Help help | The chatbot lists the supported commands/topics. |
| Unknown message I like pizza | A useful fallback message is displayed. |
| Exit bye | A goodbye message is displayed and the loop ends. |
| Exit exit | A goodbye message is displayed and the loop ends. |

## 6. Sample Demonstration

```
Welcome to the Rule-Based Python Chatbot!

What is your name? K. Bhargava Ram

Nice to meet you, K. Bhargava Ram!

You: HELLO

Hello, K. Bhargava Ram! How can I help you?

You: What is Python?

Python is a beginner-friendly programming language used for web development, automation, data analysis,
and AI.

You: hello bot

Hello, K. Bhargava Ram! How can I help you?

You: I like pizza

Sorry, I do not understand that yet. Try saying hello, asking about Python or weather, asking for help,
or typing 'bye' to exit.

You: bye

Goodbye, K. Bhargava Ram! Have a great day.
```

## 7. Why This Solution Meets the Task

This solution follows the task sheet rather than relying on a copied chatbot example. It uses the requested conversational loop, input cleaning, multiple greetings, multiple topics, if/elif/else decision making, an exit condition with break, the user's name, and a useful unknown-message response. It also implements the challenge by using the in operator so greeting variations such as “hello bot” and “hey hello” are recognized.

## 8. Final Target Flow

**INPUT → PROCESS → RESPOND → REPEAT → EXIT**

The task sheet specifically identifies this as the target flow and asks the student to test greetings, uppercase input, questions, an unknown message, and bye/exit before preparing the demonstration.